

Fairness Is Advantage-Neutral: Guided Quantum Gibbs Dynamics for Fair Allocation of Indivisible Resources

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Abstract

Fair allocation of scarce indivisible resources is a natural target for quantum optimisation, but fairness criteria such as max-min welfare and the Gini coefficient cannot be written as quadratic binary objectives without auxiliary variables. We show that this obstruction disappears once allocations are randomised. By convex duality every concave fairness criterion reduces to a weighted-welfare problem whose Hamiltonian has the degree of the valuations, at zero auxiliary cost, and ex-ante fair allocation is polynomially equivalent to weighted-welfare allocation on any computational model. Fairness is therefore advantage-neutral: a quantum speedup for fair allocation exists exactly when one exists for the underlying combinatorial problem. Repeated fair allocation becomes a multiplicative-weights game whose rounds are Gibbs sampling problems with slowly varying targets. We prove a regret guarantee and a warm-start theorem under which each round costs a number of chain steps set by the spectral gap alone. For fair scheduling under interference we construct a quantum-enhanced Metropolis sampler whose proposals come from a palindromic Trotterisation of a cost Hamiltonian plus the blockade-preserving PXP driver. We prove that the circuit never leaves the feasible set, that the proposal is exactly symmetric at every Trotter depth, and that the chain converges to the fair Gibbs lottery. We give the circuit at gate level with closed-form resource counts and verify it against exact simulation to 10^{-15} . On three graph families with up to twenty links, quantum proposals enlarge the spectral gap over local moves by factors of 3.4 to over 1000, with geometric means of 20 to 49 per family. On Erdős–Rényi graphs the fitted gap-scaling exponent improves on the best classical baseline by a factor of 1.9 ± 0.5 to 2.3 ± 0.8 , but not on unit-disk or random 3-regular graphs. In repeated max-min allocation the quantum-mixed chain reaches a fairness value of 1.327 ± 0.007 against 1.179 ± 0.139 classically, with ex-ante optimum 1.460. These results establish constant-factor and instance-dependent gains, not a general quantum advantage.

Keywords: fair division; quantum optimisation; Markov chain Monte Carlo; convex duality; independent sets; Rydberg arrays

1 Introduction

Consider a small example. Alice and Bob share three radio transmitters along a corridor (Figure 1). Neighbouring transmitters interfere: a and b cannot transmit at the same time, and neither can b and c , but a and c are far enough apart. Alice owns a and c , Bob owns b , and every transmitter sends one packet when it fires. In every time slot a scheduler must decide which transmitters fire.

Only two choices are worth making. Firing a and c gives Alice two packets and Bob none. Firing b gives Bob one packet and Alice none. Every choice therefore leaves someone with nothing. A scheduler that maximises the total always picks $\{a, c\}$, and Bob is never served.

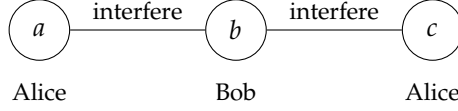


Figure 1: Three transmitters along a corridor. Neighbours interfere, so a and b cannot transmit together, nor can b and c ; a and c can. Alice owns a and c , Bob owns b .

The way out is to randomise. If the scheduler fires $\{a, c\}$ in one third of the slots and $\{b\}$ in the other two thirds, Alice expects $2 \cdot \frac{1}{3} = \frac{2}{3}$ of a packet per slot and Bob expects $1 \cdot \frac{2}{3} = \frac{2}{3}$. No rule does better for whoever is worse off, since raising either share lowers the other. Lotteries of this kind are already used to ration scarce medical treatments [89, 118].

The scheduler does not need to know the split $\frac{1}{3} : \frac{2}{3}$ in advance. It keeps a priority for Alice and one for Bob, fires the schedule with the larger priority-weighted value, and afterwards lowers the priority of whoever was just served. The priorities settle near $(\frac{1}{3}, \frac{2}{3})$, which are exactly the weights at which the two schedules tie, because $\frac{1}{3} \cdot 2 = \frac{2}{3} \cdot 1$. Two things are worth noticing. Fairness entered only through the priorities. And the only question the scheduler ever answered was the ordinary one: which schedule is best at these priorities?

With thousands of transmitters, that question becomes the maximum-weight independent set problem, which is NP-hard [43]. So instead of finding the best schedule, the scheduler draws a good one at random with a random walk that switches one transmitter on or off at a time. Here the walk is slow. To get from $\{a, c\}$ to $\{b\}$ it must switch a off, then c off, then b on, passing through schedules that are worse for everyone.

Now replace the transmitters by atoms, one per transmitter, held in a Rydberg array. Two excited atoms that are too close cannot coexist (the Rydberg blockade [12, 57]), so if the atoms are placed like the transmitters, the physics itself forbids every schedule that interferes. Let the atoms evolve for a short time and then measure them: the result is a proposed schedule, and it can jump from $\{a, c\}$ to $\{b\}$ in one move. A classical coin then accepts or rejects the proposal so that, in the long run, each schedule is used exactly as often as the fair lottery requires [69, 79].

What did the quantum device contribute? Not fairness: the priorities took care of that. It made the random walk faster. This paper shows that this is true in general. Once lotteries are allowed, fairness never makes allocation harder, and the only place a quantum device can help a fair allocator is in finding or sampling good schedules.

The example is small, but its structure is general: allocating food, vaccines, ventilators, spectrum or transmission slots under scarcity is an equity problem as much as an efficiency problem [13, 17, 82, 89]. Welfare economics offers many ways of stating what a fair allocation is, from Rawlsian max-min and leximin [93] through the Gini and Atkinson families [6, 97, 117] to Nash welfare [63, 84] and envy-based criteria [40, 113]. Quantum optimisation hardware accepts a much narrower input: a quadratic objective over binary variables, or its Ising equivalent [35, 61, 74]. Max-min welfare has multilinear degree up to the number of agents, and absolute differences, and hence the Gini coefficient, have degree at least three (Section 4). Neither can be written as a quadratic binary objective without auxiliary variables and penalties [14]. Published quantum treatments of equity therefore either substitute a quadratic surrogate for the intended criterion [112] or move the fairness objective out of the Hamiltonian altogether [10].

As the example shows, when goods are indivisible the object of fair allocation is a lottery over allocations rather than a single allocation [7, 31, 51], and randomised rationing is already used for scarce medical resources [89, 118]. Over lotteries the attainable utility set is convex. Convex duality [94] then separates the normative content of a fairness criterion from the combinatorial content of the feasible set, and this separation settles both the representa-

tion question and the question of where a quantum device can help.

Contributions.

1. A normative–combinatorial separation theorem (Theorem 1). Every concave fairness criterion, together with any requirement linear in the lottery, reduces to weighted-welfare best responses at zero auxiliary cost.
2. Advantage neutrality (Theorem 2): ex-ante fair allocation and weighted-welfare allocation are polynomially equivalent on every computational model.
3. A dynamic formulation of repeated fair allocation with a regret guarantee (Theorem 3) and a warm-start theorem (Theorem 4) that bounds the per-round sampling cost by the spectral gap, with no dependence on the size of the state space.
4. A quantum proposal for fair scheduling under interference that preserves feasibility exactly and is exactly symmetric at every Trotter depth, yielding a chain with the fair Gibbs lottery as its unique stationary distribution (Theorem 5). The circuit is given at gate level with closed-form counts.
5. An open, verified implementation and an experimental study on three graph families with five classical and quantum kernels, including exact warm-start tracking costs along multiplicative-weights trajectories.

The results are deliberately scoped. The theory shows that fairness cannot be the source of a quantum advantage. The experiments show where constraint-native quantum proposals help a fair sampler and where they do not. On present evidence, no general quantum advantage is claimed.

2 Related work

Four lines of work meet in this paper. Each has solved one part of the problem and left another open, and the gap they leave together is the one we address.

What fairness means, and why lotteries. Inequality measurement starts from the transfer principle [28] and the equally-distributed-equivalent income of Atkinson [6]; Sen and Weymark turned the Gini coefficient into a welfare function [97, 117], and the generalized Lorenz order is the consensus of all Schur-concave criteria [77, 98]. Between the utilitarian sum [51] and Rawlsian max-min [93] lie Nash welfare [63, 84], α -fairness [80], Kalai–Smorodinsky [62] and ordered weighted averages [122]; which of these is admissible depends on the informational basis of the comparison [29, 48], and each has an efficiency price [13]. For indivisible goods, exact fairness generally fails: envy-freeness up to one good exists [19, 72], maximin shares need not [67], and envy-freeness up to any good fails for submodular valuations [1]. The classical remedy, anticipated in the Harsanyi–Diamond exchange [31, 51], is to randomise: lotteries recover exact ex-ante fairness with approximate ex-post guarantees [7], although ex-ante envy-freeness with Pareto optimality is PPAD-hard [21] and variance-based objectives are NP-hard [8]. Weighted lotteries are now used to ration scarce therapeutics [89, 118]. Most relevant here, Kawase and Sumita [66] showed that a max-min fair lottery can be computed by repeatedly maximising a weighted utilitarian welfare. *Left open:* a single statement covering every concave criterion, and what it implies for the computational model on which the weighted problem is solved.

Fairness as prices on a feasible set. Convex duality [37, 94, 100] turns a concave objective over a polytope into a sequence of linear ones, and convex optimisation over a polytope reduces in polynomial time to linear optimisation over it [45]. Multiplicative weights [5, 41, 42] realise the dual as a repeated game with regret guarantees [22]. Wireless networking arrived at the same structure independently: max-weight scheduling solves a weighted independent-set problem in every slot [109], α -fair congestion control sets the weights [80], adaptive CSMA replaces the maximisation by Gibbs sampling over independent sets with dual weights [59], and the network adiabatic theorem shows that the sampler tracks slowly varying weights [92]. *Left open:* all of these samplers use local (Glauber) moves, and local chains on independent sets mix exponentially slowly at high weights, where sampling is provably hard [81, 101]. The kernel, not the fairness rule, is the bottleneck.

Quantum optimisation, and why fairness has resisted it. Quantum annealers and variational circuits take a quadratic binary objective, or its Ising form [61, 74]. The quantum approximate optimisation algorithm [35] alternates cost and mixing Hamiltonians; constraint-preserving mixers remove equality penalties [46, 115]; low-depth circuits face locality and symmetry obstructions [18, 36]; and quadratic speedups are unlikely to survive error-correction overheads [9]. Larger speedups need structure: algebraic structure in decoded quantum interferometry [4, 60], planted structure [96], or sign-problem-free adiabatic paths [44, 52]. Maximum independent set is native to Rydberg arrays through the blockade mechanism [12, 33, 57, 75, 90], but classical solvers remain competitive on unit-disk instances [3], general graphs need embedding gadgets [85], site weights have been demonstrated only at small scale [30], adiabatic gaps on hard instances can close superexponentially [95], and the strongest recent speedup claims rely on non-stoquastic drivers [23]. Fairness criteria are not quadratic, so quantum treatments of equity either substitute a quadratic surrogate [112] or remove the criterion from the Hamiltonian [10]. *Left open:* whether the representation barrier is real, and whether fairness can ever be the source of a quantum advantage.

Quantum sampling. Quantum walks improve spectral gaps quadratically [102, 108], and dynamic Gibbs sampling underlies quantum speedups for zero-sum games [15]. Quantum-enhanced Markov chain Monte Carlo keeps the classical Metropolis–Hastings filter [53, 79] and replaces only the proposal by measured quantum evolution [69]; it has since been driven by QAOA-type circuits [83], scaled beyond device size [39], combined with parallel tempering for independent-set problems [78], and partly reproduced by quantum-inspired classical proposals [24]. Its limits are also known: unital proposals give no speedup on unstructured problems [88], and there are barriers to efficient mixing [87]. Closest to our instance class, Wild et al. [119, 120] prepare coherent encodings of weighted independent-set Gibbs states on Rydberg arrays, and Cain et al. [20] prove quadratic speedups over classical Markov chains on flat landscapes. *Left open:* a sampler whose correctness does not depend on adiabaticity or noise-free hardware, whose target moves with the fairness weights, and whose proposal never leaves the feasible set.

Position of this paper. Table 1 summarises the comparison. The separation theorem generalises the max-min reduction of [66] to every concave criterion and turns it into a statement about quantum advantage; the warm-start theorem is an explicit, $|F|$ -independent form of the adiabatic tracking of [92]; and the quantum proposal replaces the local moves of adaptive CSMA [59] with blockade-preserving evolution inside an exact Metropolis filter [69]. To our knowledge, a quantum proposal has not previously been used inside a fairness-driven Gibbs sampler.

	all concave criteria	moving Gibbs target	$ F $ -free tracking	quantum proposal	exact feasibility & stationarity
Kawase–Sumita [66]	max-min only	–	–	–	n/a
Adaptive CSMA [59]	utility max.	✓	–	–	✓
Network adiabatic [92]	–	✓	asymptotic	–	✓
QeMCMC [69]	–	–	–	✓	unconstrained
Rydberg Gibbs states [120]	–	–	–	adiabatic	ideal hardware
This work	✓	✓	✓	✓	✓

Table 1: Closest prior work. “Exact feasibility and stationarity”: the sampler never leaves F and has the target as its exact stationary distribution.

3 Problem formulation

Allocations and utilities. There are n agents, indexed by $i = 1, \dots, n$, and a finite set F of feasible allocations. Each allocation is a bit string $x = (x_1, \dots, x_M) \in \{0, 1\}^M$, where $x_v = 1$ means that resource v is used. Allocation x gives agent i the utility $u_i(x) \geq 0$. We collect the utilities into the vector $u(x) = (u_1(x), \dots, u_n(x))$ and write $\rho = \max_{i,x} u_i(x)$ for the largest utility any agent can receive; ρ sets the scale of every error bound below.

Lotteries. A lottery p is a probability distribution on F : $p(x) \geq 0$ and $\sum_{x \in F} p(x) = 1$. Its expected utility vector is

$$y(p) = \sum_{x \in F} p(x) u(x), \quad (1)$$

so that $y_i(p)$ is the utility agent i expects before the lottery is drawn (ex-ante utility [7]). Because (1) is an average of the points $u(x)$ with weights $p(x)$, the set of all achievable vectors $y(p)$ is exactly the convex hull P of those points [94]. This is the reason lotteries help: P is convex even though F is a discrete set.

Fairness criteria. A fairness criterion is a concave function $W : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{-\infty\}$ of the utility vector. Concavity means that spreading utility more evenly never lowers W , which is the transfer principle of inequality measurement [6, 28]. Table 2 lists the standard criteria. The fair allocation problem is to find a lottery whose value $W(y(p))$ is within ε of the best possible value

$$W_{\max} = \max_p W(y(p)), \quad (2)$$

where the maximum runs over all lotteries p on F .

Best weighted welfare. Give each agent a priority weight $\theta_i \geq 0$ and write $\theta \cdot u(x) = \sum_i \theta_i u_i(x)$ for the weighted welfare of allocation x . The best weighted welfare is

$$B(\theta) = \max_{x \in F} \theta \cdot u(x). \quad (3)$$

The weights play the role of Lagrange multipliers, or prices, on the agents [16]. Computing $B(\theta)$ is an ordinary combinatorial problem that involves no fairness at all; for the scheduling problem of Section 6 it is maximum-weight independent set [109]. In convex analysis B is the support function of P , the largest value of the linear function $\theta \cdot y$ over P [94].

Criterion	$W(y)$	Allowed weights θ	Ref.
Utilitarian	$\sum_i y_i$	$\theta = \mathbf{1}$	[51]
Max-min	$\min_i y_i$	probability vectors	[93]
Leximin	max-min, then the next-worst, ...	nested max-min stages	[29]
Lorenz component	sum of the k smallest y_i	$0 \leq \theta_i \leq 1, \sum_i \theta_i = k$	[98]
Ordered weighted average	$\sum_k \omega_k y_{(k)}, \omega$ decreasing	mixtures of permutations of ω	[122]
Gini–Sen welfare	mean $\times (1 - \text{Gini})$	permutations of Gini weights	[97, 117]
Nash	$\sum_i \log y_i$	$\theta_i = 1/y_i$ at the optimum	[63, 84]
α -fair, Atkinson	$\sum_i y_i^{1-\alpha} / (1 - \alpha)$	$\theta_i = y_i^{-\alpha}$ at the optimum	[6, 80]
Kalai–Smorodinsky	$\min_i (y_i - d_i) / (a_i - d_i)$	rescaled probability vectors	[62]
Need floors, proportionality	$y_i \geq d_i, y_i \geq \frac{1}{n} \sum_j y_j$	one multiplier per constraint	[104]
Ex-ante envy-freeness	$\mathbb{E}u_i(x_i) \geq \mathbb{E}u_i(x_j)$	one multiplier per pair	[40, 113]
Mean minus variance	$\text{mean}(y) - \lambda \text{Var}(y)$	weights can be negative	[76]

Table 2: Fairness criteria and the priority weights they allow. $y_{(k)}$ is the k -th smallest utility.

Conjugate of the criterion. The criterion enters only through its concave conjugate (the Legendre–Fenchel transform [16, 37, 94]),

$$W^*(\theta) = \inf_y [\theta \cdot y - W(y)]. \quad (4)$$

For a given θ , $W^*(\theta)$ is the largest constant c such that the linear function $\theta \cdot y - c$ lies above W everywhere. We write $\inf$ rather than $\min$ because the value can be $-\infty$; it is finite only on the weights that W allows. For max-min fairness, $W(y) = \min_i y_i$: the allowed weights are the probability vectors ($\theta_i \geq 0, \sum_i \theta_i = 1$), and $W^* = 0$ on them. The last column of Table 2 gives the allowed weights of every criterion; Appendix A derives them.

4 Fairness is a choice of prices

Theorem 1 (Separation). *For every concave fairness criterion W ,*

$$\max_p W(y(p)) = \inf_{\theta} [B(\theta) - W^*(\theta)]. \quad (5)$$

The left side is the fair allocation problem (2). The right side is its dual: for each choice of weights θ , the number $B(\theta) - W^*(\theta)$ is an upper bound on the achievable fairness, and the best such bound is exact. The right side splits the problem in two. The *normative* part, W^* , says which weights are allowed and at what cost; it depends only on the criterion. The *combinatorial* part, B , only ever asks for the best allocation at given weights; it depends only on F and u .

Proof. By (1) the left side is the maximum of W over the polytope P . A closed concave function is the pointwise minimum of the linear functions that lie above it, so $W(y) = \inf_{\theta} [\theta \cdot y - W^*(\theta)]$ (Fenchel–Moreau theorem [37, 94]). The bracket is linear in y and convex in θ , and P is compact, so the maximum over y and the infimum over θ can be exchanged (Sion’s minimax theorem [100]). A linear function is maximised over a polytope at a vertex, and the vertices of P are points $u(x)$, so $\max_{y \in P} \theta \cdot y = B(\theta)$. $\square$

No representation cost. The problem $B(\theta)$ in (3) is linear in the utilities, so it has the same polynomial degree in the bits x_v as the utilities themselves, whatever the criterion. The criteria themselves, evaluated on a single allocation, can have much higher degree. With $u_i = x_i$, max-min is $\min_i u_i = x_1 x_2 \cdots x_n$, of degree n , and the absolute differences in the Gini coefficient have degree at least three [14]. Neither appears in B , so no criterion in Table 2 needs auxiliary variables or penalty terms.

Theorem 2 (Advantage neutrality). *On any computational model, classical or quantum, fair allocation and weighted-welfare allocation over the same feasible set are polynomially equivalent.*

“Polynomially equivalent” means that each problem can be solved by a polynomial number of calls to a solver for the other, with only polynomial extra work [43].

Proof. A first-order method on the right side of (5) reaches accuracy ε after a number of steps polynomial in n , ρ and $1/\varepsilon$, and each step needs one approximate evaluation of B [5, 42]; for max-min this is the reduction of Kawase and Sumita [66]. Conversely, a linear criterion $W(y) = \theta \cdot y$ is itself a fair allocation problem, and its optimum is attained by a single allocation, which solves $B(\theta)$. $\square$

Fairness therefore cannot be the source of a quantum speedup. A quantum device can help only with the weighted problem B , or with sampling good solutions of it.

Where fairness itself is hard. Deterministic max-min allocation is NP-hard already for two agents with identical additive valuations, by reduction from PARTITION [43, 64]. The lottery version of the same instance is solved by a fair coin. For additive valuations the lottery optimum can be rounded to a deterministic allocation that loses at most n items’ worth of utility per agent, by the Shapley–Folkman theorem [103].

5 Repeated fair allocation

Weight update. Allocations are made in rounds $t = 1, \dots, T$; x_t is the allocation of round t and θ_t the weights used to choose it. The weights start equal and are updated by the Hedge (multiplicative weights) rule [5, 41, 42],

$$\theta_{t+1,i} = \frac{\theta_{t,i} e^{-\eta u_i(x_t)}}{\sum_{j=1}^n \theta_{t,j} e^{-\eta u_j(x_t)}}, \quad \theta_{1,i} = \frac{1}{n}. \quad (6)$$

Here $\eta > 0$ is the step size and the denominator keeps the weights summing to one. Each weight is multiplied by $e^{-\eta u_i(x_t)}$, so an agent who received a lot loses priority and an agent who received little gains it. The exponential form is the one that gives the regret bound used in Theorem 3 [22, 41]. In game terms, max-min fairness is a zero-sum game between an allocator, who picks x , and an adversary, who picks the agent i to be judged; (6) is the adversary learning which agent is worst off [42].

Gibbs sampling. Instead of solving $B(\theta_t)$ exactly, each round draws its allocation from the Gibbs (Boltzmann) distribution

$$\pi_t(x) = \frac{e^{\beta n \theta_t \cdot u(x)}}{Z_t}, \quad Z_t = \sum_{x' \in F} e^{\beta n \theta_t \cdot u(x')}. \quad (7)$$

Z_t is the normalising constant (the partition function), and $\beta \geq 0$ is the inverse temperature of statistical physics [58, 79]: $\beta = 0$ gives the uniform lottery on F , and $\beta \rightarrow \infty$ concentrates on the maximisers of $B(\theta_t)$. The factor n compensates for the weights summing to one, so that the average weight per agent, $n\theta_{t,i}$, is of order one for every n . Among all lotteries with the same expected weighted welfare, the Gibbs distribution has the largest entropy, so it commits to nothing beyond what the weights require [58]. The price of sampling instead of maximising is known exactly:

$$\mathbb{E}_{\pi_t}[\theta_t \cdot u] \geq B(\theta_t) - \frac{\log |F|}{\beta n}, \quad (8)$$

where $\mathbb{E}_{\pi_t}$ is the expectation over $x \sim \pi_t$ and $|F|$ the number of allocations. It holds because π_t maximises $\beta n \mathbb{E}[\theta_t \cdot u] + H$, where H is the Shannon entropy, and $0 \leq H \leq \log |F|$ [26].

Theorem 3 (Fairness guarantee). *Let v^* be the value of (2) for max-min fairness, and $\eta = \rho^{-1} \sqrt{8 \ln n / T}$. If each x_t is drawn within total variation distance ε of π_t , the worst-off agent's average utility satisfies*

$$\mathbb{E} \left[\min_i \frac{1}{T} \sum_{t=1}^T u_i(x_t) \right] \geq v^* - \frac{\log |F|}{\beta n} - \rho \varepsilon - \rho \sqrt{\frac{\ln n}{2T}}. \quad (9)$$

The total variation distance between two distributions μ and π is $\frac{1}{2} \sum_x |\mu(x) - \pi(x)|$ [71]. The left side of (9) is the long-run utility of the worst-off agent. The three losses on the right are, in order, the price of sampling instead of maximising (8), the sampling error, and the regret of learning the weights; the last vanishes as $T \rightarrow \infty$. The value of η is the one that minimises the regret term [22].

Proof. By (5) with $W = \min_i v^* = \min_{\theta} B(\theta) \leq B(\theta_t)$ for every t . By (8) the expected weighted welfare in round t is at least $B(\theta_t) - \log |F| / (\beta n)$, and a sampling error ε changes any expectation of a quantity bounded by ρ by at most $\rho \varepsilon$. The Hedge regret bound, proved with Hoeffding's lemma [22, 54], gives $\sum_t \theta_t \cdot u(x_t) - \min_i \sum_t u_i(x_t) \leq \rho \sqrt{T \ln n / 2}$. Dividing by T gives (9). $\square$

The target moves slowly. By (6) each weight changes by a factor of at most $e^{\pm \eta \rho}$ per round. Let $\delta_t = \max_x |(\theta_{t+1} - \theta_t) \cdot u(x)|$ be the largest change in weighted welfare between rounds. Then, for every allocation x ,

$$e^{-2\beta n \delta_t} \leq \frac{\pi_{t+1}(x)}{\pi_t(x)} \leq e^{2\beta n \delta_t}. \quad (10)$$

The exponent is $2\beta n \delta_t$ rather than $\beta n \delta_t$ because both the numerator of (7) and its normalising constant Z_t change, each by a factor of at most $e^{\pm \beta n \delta_t}$. A sampler that is already close to π_t therefore needs only a few steps to reach π_{t+1} .

Theorem 4 (Warm start). *Let the Markov chain used in round $t + 1$ be reversible with stationary distribution π_{t+1} , non-negative spectrum and spectral gap γ . If it starts from a distribution μ with $\chi^2(\mu \| \pi_t) \leq \varepsilon^2$, then after*

$$k \geq \frac{1}{2\gamma} \ln \frac{e^{2\beta n \delta_t} (1 + \varepsilon^2) - 1}{\varepsilon^2} \quad (11)$$

steps its distribution is within χ^2 -distance ε^2 of π_{t+1} . The bound does not depend on the number of allocations $|F|$.

The χ^2 -distance is $\chi^2(\mu \| \pi) = \sum_x \mu(x)^2 / \pi(x) - 1$; it bounds the total variation distance by $\text{TV} \leq \frac{1}{2} \sqrt{\chi^2}$ [71]. A chain is reversible if $\pi(x)P(x, y) = \pi(y)P(y, x)$ for all x, y , where $P(x, y)$ is the probability of moving from x to y ; its spectral gap is $\gamma = 1 - \lambda_2$, where λ_2 is the second-largest eigenvalue of P [71]. Non-negative spectrum, which the lazy chains of Section 6 have, makes γ also the absolute gap that controls convergence. Roughly $1/\gamma$ steps are needed to forget the starting point. The numerator in (11) is the distance to the new target just after the weights change, and the factor $1/(2\gamma)$ is the time needed to shrink it to ε^2 .

Proof. By (10), $1 + \chi^2(\mu \| \pi_{t+1}) \leq e^{2\beta n \delta_t} (1 + \chi^2(\mu \| \pi_t))$, so the distance to the new target is at most $e^{2\beta n \delta_t} (1 + \varepsilon^2) - 1$. Each step of a reversible chain multiplies the χ^2 -distance by at most $(1 - \gamma)^2 \leq e^{-2\gamma}$, because $\mu/\pi - 1$ is orthogonal to the constant functions [71]. Solving $e^{-2\gamma k} (e^{2\beta n \delta_t} (1 + \varepsilon^2) - 1) \leq \varepsilon^2$ for k gives (11). $\square$

Theorem 4 is an explicit form of the network adiabatic theorem [92]. Together with Theorem 3, it shows that the cost of fair allocation is set by one number, the spectral gap γ of the sampler. That is the quantity a quantum proposal can change [69].

6 The quantum proposal

6.1 Fair scheduling under interference

N wireless links share a conflict graph [56]: two links joined by an edge interfere and cannot transmit together. We write $N(v)$ for the neighbours of link v and $d_v = |N(v)|$ for its degree. A schedule is an independent set x of the graph: $x_v = 1$ if link v transmits, and no two neighbours both transmit. Link v has rate $r_v > 0$ and belongs to agent $a(v)$, so agent i receives $u_i(x) = \sum_{v: a(v)=i} r_v x_v$. Substituting this into (7) gives

$$\pi(x) \propto e^{\beta \sum_v w_v x_v}, \quad w_v = n \theta_{a(v)} r_v, \quad (12)$$

where $\propto$ means “proportional to”, the constant being $1/Z_t$. The link weight w_v is the rate of the link times the priority of its owner, with the same factor n as in (7). For this target, B is the maximum-weight independent set problem that max-weight scheduling solves in every slot [109]. Adaptive CSMA samples the same target with single-link moves [59]. We keep its Metropolis filter and replace the moves.

6.2 Hamiltonian

We use Dirac notation [86]: each link is a qubit, $|0\rangle$ means “off” and $|1\rangle$ means “on”, and a schedule x is the basis state $|x\rangle = |x_1\rangle \cdots |x_N\rangle$. Three operators on link v are needed. $n_v = |1\rangle\langle 1|_v$ equals 1 if link v is on and 0 if it is off. X_v is the Pauli-X (bit-flip) operator, which swaps $|0\rangle_v$ and $|1\rangle_v$. And $\Pi_v = \prod_{u \in N(v)} |0\rangle\langle 0|_u$ projects onto states in which every neighbour of v is off. The proposal evolves under

$$H = \underbrace{-\gamma_c \sum_v w_v n_v}_{\text{cost}} + \underbrace{h \sum_v X_v \Pi_v}_{\text{driver}}, \quad (13)$$

with positive strengths γ_c and h . The cost term is diagonal: on $|x\rangle$ it takes the value $-\gamma_c \sum_v w_v x_v$, so valuable schedules have low energy, as in the cost Hamiltonians of quantum optimisation [35, 74]. The driver is the PXP model of Rydberg-blockaded atoms [38, 70, 111]; its name reads $\Pi_v - X_v - \Pi_v$ along a chain. The product $X_v \Pi_v$ flips link v only when all its neighbours are off, which is exactly the rule that keeps a schedule independent. On a Rydberg array, (13) is the native Hamiltonian, with local detuning $\gamma_c w_v$ and Rabi frequency $2h$ [12, 90].

6.3 Palindromic Trotter step

The two terms of (13) do not commute, so the evolution e^{-iHt} is approximated by alternating short evolutions under each term, a Trotter splitting [73, 110]. Two driver terms $X_u \Pi_u$ and $X_v \Pi_v$ on non-adjacent links do commute, because each acts on sites where the other is the identity. We therefore colour the conflict graph properly with colours $C_1, \dots, C_\chi$, so that no two links of the same colour are adjacent [116], and define one exact layer per colour and one cost layer:

$$D_C(s) = \prod_{v \in C_C} e^{-is h X_v \Pi_v}, \quad C(s) = e^{is \gamma_c \sum_v w_v n_v}. \quad (14)$$

$D_c(s)$ evolves every link of colour c under the driver for time s , and $C(s)$ evolves all links under the cost term for time s . With K steps of length $\tau = t_{\text{ev}}/K$ for a total evolution time t_{ev} , one step is

$$S(\tau) = C(\frac{\tau}{2}) D_1(\frac{\tau}{2}) \cdots D_{\chi-1}(\frac{\tau}{2}) D_\chi(\tau) D_{\chi-1}(\frac{\tau}{2}) \cdots D_1(\frac{\tau}{2}) C(\frac{\tau}{2}). \quad (15)$$

This is the symmetric (Strang) splitting [105, 106]. For two terms A and B , $e^{-i\tau(A+B)} \approx e^{-i\tau A/2} e^{-i\tau B} e^{-i\tau A/2}$, with error of order τ^3 per step instead of τ^2 for the one-sided product $e^{-i\tau A} e^{-i\tau B}$. Applying the same rule to all $\chi + 1$ layers in turn gives (15). Every layer runs for $\frac{\tau}{2}$ on each side, except the middle layer D_χ , whose two halves are adjacent and merge into one layer of length τ . The result reads the same forwards and backwards (a palindrome), and this, not the accuracy, is what the proof of exactness uses. The proposal unitary is

$$U = S(\tau)^K \approx e^{-iHt_{\text{ev}}}, \quad (16)$$

with total error of order $t_{\text{ev}}\tau^2$ [106]. To propose a move from the current schedule y , prepare $|y\rangle$, apply U and measure every qubit. By the Born rule [86] the result is x with probability

$$Q(x | y) = |\langle x | U | y \rangle|^2, \quad (17)$$

where $\langle x | U | y \rangle$ is the amplitude of $|x\rangle$ in the state $U | y \rangle$, and $Q(x | y)$ is read “the probability of proposing x given the current schedule y ”. This construction follows quantum-enhanced Markov chain Monte Carlo [69], with the PXP driver in place of a transverse field.

6.4 Exactness

Three facts, each one line, make the sampler exact.

1. *It never leaves the feasible set.* The cost term is diagonal, and $X_v \Pi_v$ acts only when all neighbours of v are off, so every layer in (14) maps independent sets to independent sets. Hence $Q(x | y) = 0$ whenever x is not an independent set.
2. *It is exactly symmetric.* Every factor in (15) is the exponential of i times a real symmetric matrix, so it equals its own transpose, and the transpose of a palindrome is the same palindrome. Hence $U^\top = U$ and $Q(x | y) = |\langle x | U | y \rangle|^2 = |\langle y | U | x \rangle|^2 = Q(y | x)$ for every K and τ .
3. *A symmetric proposal inside Metropolis is exact.* If a symmetric proposal is accepted with probability $\min\{1, \pi(x)/\pi(y)\}$, the chain satisfies detailed balance with respect to π [53, 79].

For the target (12), the acceptance ratio is $\pi(x)/\pi(y) = e^{\beta \sum_v w_v(x_v - y_v)}$; the normalising constant cancels, so it never has to be computed.

Theorem 5 (Correctness). *Mix the quantum proposal with single-link flips, using the quantum move with probability $q < 1$, accept a proposed x from y with probability $\min\{1, \pi(x)/\pi(y)\}$, and stay put with probability $\frac{1}{2}$ at every step. The resulting chain on the independent sets is reversible, irreducible and aperiodic, has non-negative spectrum, and has π in (12) as its unique stationary distribution.*

Proof. Let $\tilde{Q} = (1 - q)Q_{\text{flip}} + qQ$ be the mixed proposal, where Q_{flip} flips one link chosen uniformly at random. Both parts are symmetric, so for $x \neq y$ the transition probability $P(y, x) = \frac{1}{2}\tilde{Q}(x | y) \min\{1, \pi(x)/\pi(y)\}$ satisfies $\pi(y)P(y, x) = \frac{1}{2}\tilde{Q}(x | y) \min\{\pi(x), \pi(y)\}$, which is symmetric in x and y (detailed balance). Single flips connect every independent set to the empty set, so the chain is irreducible. Staying put with probability $\frac{1}{2}$ gives aperiodicity and eigenvalues in $[0, 1]$ [71]. $\square$

6.5 Circuit

Every gate follows from three identities [11, 86]:

$$e^{is\gamma_c w_v n_v} = P(s\gamma_c w_v), \quad P(\phi) = \text{diag}(1, e^{i\phi}), \quad (18)$$

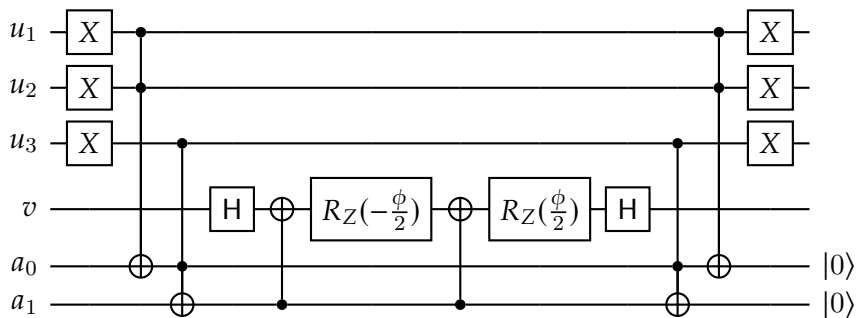
$$e^{-i\alpha X_v \Pi_v} = R_X(2\alpha) \text{ on } v, \text{ controlled on every neighbour being } |0\rangle, \quad (19)$$

$$CR_X(\phi) = (I \otimes H)(I \otimes R_Z(\frac{\phi}{2})) \text{CNOT}(I \otimes R_Z(-\frac{\phi}{2})) \text{CNOT}(I \otimes H). \quad (20)$$

Here $P(\phi)$ is the phase gate; $R_X(\phi) = e^{-i\phi X/2}$ and $R_Z(\phi) = e^{-i\phi Z/2}$ are rotations about the x and z axes; CR_X is R_X controlled by one qubit; CNOT is the controlled-NOT; and H is the Hadamard gate, in sans-serif to distinguish it from the Hamiltonian H . Each identity, and each choice in the circuit below, is a standard construction:

1. *Cost layer* (18). n_v is diagonal with eigenvalues 0 and 1, so $e^{is\gamma_c w_v n_v} = \text{diag}(1, e^{is\gamma_c w_v})$, which is a single phase gate on link v . The cost layer needs no two-qubit gates.
2. *Driver term* (19). X_v and Π_v act on different qubits and commute, and Π_v is a projector ($\Pi_v^2 = \Pi_v$). Expanding the exponential then gives $e^{-i\alpha X_v \Pi_v} = \Pi_v \otimes R_X(2\alpha)_v + (I - \Pi_v) \otimes I_v$: the rotation is applied exactly when every neighbour is $|0\rangle$. The factor 2 comes only from the convention $R_X(\phi) = e^{-i\phi X/2}$.
3. *Controls on $|0\rangle$* . A control usually fires on $|1\rangle$. Applying X to a neighbour before and after the controlled gate makes it fire on $|0\rangle$, because $X|0\rangle\langle 0|X = |1\rangle\langle 1|$.
4. *Many controls*. Link v has d_v neighbours, so the rotation needs d_v controls. The standard construction [86, Fig. 4.10] computes the AND of the controls into $d_v - 1$ ancilla qubits with a ladder of Toffoli gates: the first ancilla holds $u_1 \wedge u_2$, the next holds that AND u_3 , and so on. The last ancilla then controls the rotation, and the ladder is run backwards to return every ancilla to $|0\rangle$, so that no ancilla stays entangled with the links.
5. *One controlled rotation from two CNOTs* (20). Any controlled single-qubit gate can be written as $A \text{CNOT} B \text{CNOT} C$ with $ABC = I$ (Barenco et al., Lemma 5.1 [11]; Corollary 4.2 of [86]). Here $C = H$, $B = R_Z(-\frac{\phi}{2})$ and $A = HR_Z(\frac{\phi}{2})$. If the control is $|0\rangle$, the CNOTs do nothing and the product is $ABC = HR_Z(\frac{\phi}{2})R_Z(-\frac{\phi}{2})H = I$. If the control is $|1\rangle$, the CNOTs act as X on the target, and since $XR_Z(-\frac{\phi}{2})X = R_Z(\frac{\phi}{2})$ and $HR_Z(\phi)H = R_X(\phi)$, the product is $R_X(\phi)$.

This is the textbook construction, chosen because every step can be checked; it is not optimised for gate count, and constructions with fewer ancillas exist [11]. The whole circuit was also checked numerically: simulated gate by gate with its ancillas, it agrees with the exact subspace evolution to 3.1×10^{-15} , and leakage out of the feasible set below 2.1×10^{-14} (Section 7.2). For a link v with three neighbours u_1, u_2, u_3 and two ancillas a_0, a_1 , reading left to right: X on each neighbour; two Toffolis put $u_1 \wedge u_2 \wedge u_3$ (with every neighbour flipped) into a_1 ; the controlled rotation of item 5 acts on v ; the Toffolis and X gates are undone in reverse order.



Counting gates link by link gives closed forms. In (15) every link outside the middle colour class C_χ is driven twice per step (once on each side of the palindrome), and every link in C_χ once, so let $m_v = 1$ if $v \in C_\chi$ and $m_v = 2$ otherwise. Each application of the driver to link v uses $2d_v$ gates X (before and after on each neighbour), $2(d_v - 1)$ Toffolis (compute and uncompute) and, if $d_v \geq 1$, two CNOTs, two H and two R_Z . Each cost layer uses one phase gate per link, and there are two cost layers per step. One step therefore uses

$$2N \text{ phase gates, } \sum_v 2m_v d_v \text{ } X \text{ gates, } \sum_v 2m_v (d_v - 1)^+ \text{ Toffolis, } \sum_{v: d_v \geq 1} 2m_v \text{ CNOTs,} \quad (21)$$

with as many H and R_Z gates as CNOTs and $(z)^+ = \max(z, 0)$, on $N + \max(\Delta - 1, 0)$ qubits, where $\Delta = \max_v d_v$ is the maximum degree. For the benchmark unit-disk instance ($N = 20$, 53 edges, $\Delta = 10$, $\chi = 6$; Appendix B) with $t_{\text{ev}} = 5$, $K = 10$, $\gamma_c = 1$ and $h = 0.4$, one step has 40 phase, 400 X , 324 Toffoli, 76 CNOT, 76 Hadamard and 76 R_Z gates on 29 qubits, with depth 445. Decomposing each Toffoli into six CNOTs [2, 86] gives 2020 CNOTs per step. On a Rydberg array in the blockade regime no compilation is needed, since (13) is the native Hamiltonian [12, 90, 111]. Site-resolved weights of this kind have been demonstrated with local light shifts [30]; conflict graphs that are not unit-disk graphs need embedding gadgets [85].

6.6 Algorithm

Input. The conflict graph (which pairs of links cannot transmit together); the rate of each link and the agent who owns it; the number of rounds T ; the sampling sharpness β ; the learning rate η ; the number of sampling steps per round (k_0 in the first round, k after that); and the chance q of using a quantum move.

Output. A schedule for every round; the fairness achieved, which is the average share of the worst-off agent, $\min_i \frac{1}{T} \sum_t u_i(x_t)$; and the upper bound $\min_t B(\theta_t)$ on the best fairness possible, which holds by (5).

- 1: Colour the conflict graph so that no two neighbouring links share a colour.
- 2: Give every agent the same priority $\theta_i = 1/n$, and start with no link transmitting.
- 3: **for** each round $t = 1, \dots, T$ **do**
- 4: Give each link the weight $w_v = n \theta_{a(v)} r_v$: its rate times its owner's priority.
- 5: **for** k_t sampling steps **do**
- 6: Toss a fair coin. On heads, do nothing in this step.
- 7: Otherwise, propose a new schedule x from the current schedule y :
- 8: with chance q , run the circuit $S(\tau)^K$ on $|y\rangle$ and measure (17);
- 9: otherwise, switch one random link on or off, if the schedule stays valid.
- 10: Move to x with probability $\min\{1, e^{\beta(w(x) - w(y))}\}$, where $w(x) = \sum_v w_v x_v$.
- 11: Use the current schedule for round t .
- 12: Raise the priority of agents who received little, and lower it for agents who received a lot, by (6).

The coin toss in line 6 makes the chain lazy, which keeps its eigenvalues non-negative (Theorem 5). Line 10 is the Metropolis rule for the target (12). Because the schedule is carried over from one round to the next instead of restarting, each round starts close to its new target; this is the warm start of Theorem 4. The cost is $O((k_0 + Tk)(N + qc))$ operations, where c is the gate count of one quantum proposal: there are $k_0 + Tk$ sampling steps, a local flip costs $O(N)$, and a quantum proposal costs c .

7 Experiments

7.1 Setup

Three conflict-graph families were used:

- unit-disk graphs [25], with N points uniform in a square of area $N/2.5$, joined at distance at most one, for $N = 8, \dots, 20$;
- Erdős–Rényi graphs $G(N, 0.3)$ [34], whose mean degree grows with N , for $N = 8, \dots, 18$;
- uniform random 3-regular graphs [121], for $N = 8, \dots, 18$.

Each size used five seeds. Rates are drawn from $U[0.5, 1.5]$ and three agents own the links in balanced random order; all instances are fully specified by their seeds. The feasible sets ranged from 14 to 2635 independent sets.

Five symmetric proposals were compared, each inside the same lazy Metropolis kernel:

- *local*: single-vertex flips;
- *local+swap*: single flips mixed equally with Kawasaki-type moves [65] of a particle across an edge;
- *uniform*: uniform proposals over F , an idealised baseline, since exact uniform sampling of independent sets is intractable in general [32, 81, 91, 101];
- *quantum*: an average over three parameter draws with $h \in [0.25, 0.6]$, $t_{\text{ev}} \in [2, 10]$, $\tau \leq 0.5$ and $\gamma_c = (1 - h)\sqrt{N}/\|w\|$, following [69];
- *mixed*: local and quantum with equal probability.

Proposals were computed by exact evolution restricted to the span of the independent sets, and spectral gaps by diagonalising the symmetrised kernel [71]. Fixed-weight gaps use $w_v = r_v$ with $\beta \in \{1, 3\}$. Graphs were generated with NetworkX [47] and all linear algebra used NumPy and SciPy [50, 114].

7.2 Verification

On five instances covering all three families, the gate-level circuit, simulated as a full state vector with ancillas, agreed with the subspace evolution to at most 3.1×10^{-15} , with leakage below 2.1×10^{-14} . The closed-form gate counts matched the generated circuits exactly. Proposal asymmetry was at most 1.7×10^{-16} , detailed-balance error at most 6.5×10^{-19} , and stationarity error at most 1.1×10^{-16} .

7.3 Spectral gaps

Figure 2 shows the gaps. Quantum proposals enlarge the gap over local moves in every instance, by factors between 3.4 and 1116; the geometric means are 25.6 on unit-disk, 20.3 on Erdős–Rényi and 48.6 on 3-regular graphs. The relevant question for advantage is scaling, so we fit $\log \gamma = -\kappa N + c$ and report the ratio $a = \kappa_C / \kappa_Q$ against the best classical kernel. This ratio is the exponent in the relation (classical steps) $\approx$ (quantum steps) ^{a} [69].

Table 3 shows that the scaling behaviour depends on the family. On Erdős–Rényi graphs the quantum kernel improves the scaling rate by a factor of about two. On unit-disk graphs it improves only at high temperature. On random 3-regular graphs the swap-augmented classical chain scales better than the quantum kernel.

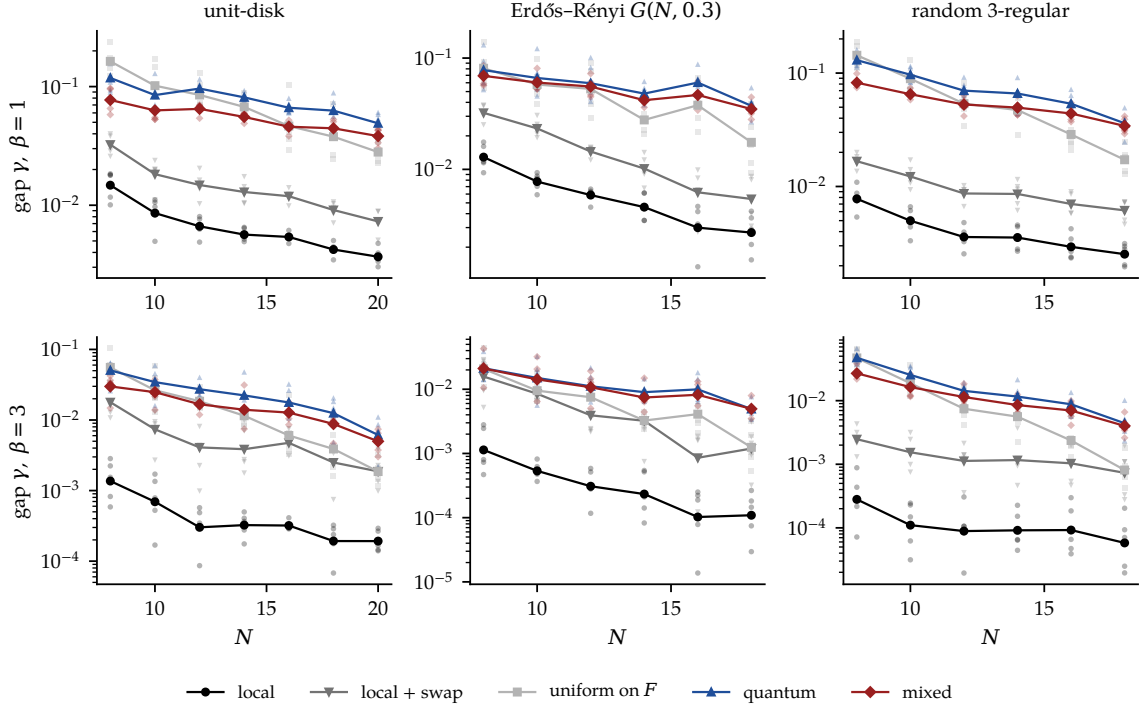


Figure 2: Spectral gaps of the five kernels against N for three graph families and two inverse temperatures. Points are individual instances and lines are geometric means over five seeds.

Family	β	κ_Q	best classical κ_C	$a = \kappa_C / \kappa_Q$
Unit-disk	1	0.065 ± 0.009	0.103 ± 0.010 (local)	1.60 ± 0.28
Unit-disk	3	0.157 ± 0.020	0.150 ± 0.026 (local)	0.95 ± 0.21
Erdős-Rényi	1	0.060 ± 0.017	0.137 ± 0.023 (uniform)	2.28 ± 0.76
Erdős-Rényi	3	0.125 ± 0.024	0.242 ± 0.042 (local)	1.94 ± 0.50
3-regular	1	0.118 ± 0.011	0.095 ± 0.011 (local+swap)	0.81 ± 0.12
3-regular	3	0.218 ± 0.020	0.104 ± 0.035 (local+swap)	0.48 ± 0.17

Table 3: Fitted gap-scaling rates with standard errors. Values of a above one favour the quantum kernel.

7.4 Tracking along fairness trajectories

Theorem 4 identifies the tracking cost as the operational quantity. For each instance we generated a multiplicative-weights trajectory with exact Gibbs sampling. For checkpoints thirty rounds apart, we computed exactly the least number of steps k with $\chi^2(\pi_t P_{t+30}^k \| \pi_{t+30}) \leq 0.01$. Figure 3 shows the results. The quantum kernel required the fewest steps in 82% of the 39 instances. In geometric mean it needed 21.4 steps against 32.3 for uniform proposals, 120.6 for local+swap and 1910 for local moves. Per instance, the best classical kernel required 1.44 times as many steps as the quantum kernel in geometric mean. The growth of the tracking cost with N is similar for the quantum and uniform kernels over the range tested.

7.5 Repeated max-min allocation

On the unit-disk instance with $N = 12$ and seed 1 ($|F| = 131$, three agents), the ex-ante optimum is $v^* = 1.460$ and the best single allocation achieves 1.227. With $\beta = 3$, $T = 1500$, $k = 8$, $k_0 = 200$ and five repetitions, Theorem 3 guarantees at least $v^* - 0.620$.

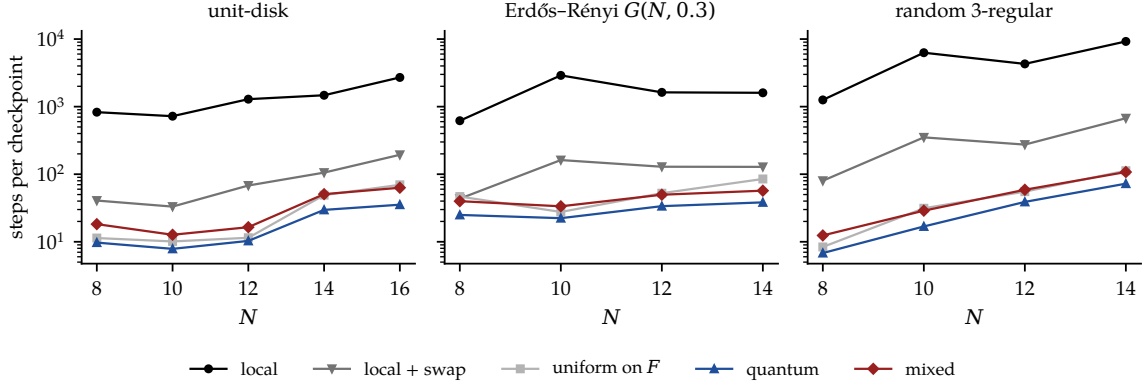


Figure 3: Exact warm-start tracking cost: the number of chain steps needed to move from π_t to within $\chi^2 \leq 0.01$ of π_{t+30} along multiplicative-weights trajectories ($T = 300, \beta = 3$). Values are geometric means over three seeds and nine checkpoints.

Chain	Fairness value	Gini of averages	Dual upper bound
classical ($q = 0$)	1.179 ± 0.139	0.0127	1.467
quantum-mixed ($q = 0.5$)	1.327 ± 0.007	0.0104	1.470

At equal step budget, the quantum-mixed chain exceeds the best deterministic allocation in every repetition and reduces the run-to-run standard deviation by a factor of about nineteen. The classical chain falls below the deterministic optimum in three of five runs.

8 Discussion

The theory fixes where a quantum device can contribute. Fairness adds no hardness *ex ante*, and no representation cost once lotteries are admitted (Theorems 1 and 2). Any speedup must therefore come from sampling the weighted feasible set, and the warm-start theorem makes the spectral gap the governing quantity.

The experiments show that constraint-native quantum proposals improve this quantity by large constant factors in all families and reduce tracking cost in most instances. They improve the scaling rate on dense random graphs but not on sparse regular ones. For a reversible kernel, the spectral gap is at most twice the conductance of any set [71, 99]. A quantum proposal therefore helps only by moving probability across the bottlenecks of π , and the family dependence observed here is consistent with the absence of a generic mechanism [87, 88] and with the finding that part of the quantum proposal’s benefit can be reproduced by quantum-inspired classical moves [24].

The Erdős–Rényi ratios of 1.9 ± 0.5 and 2.3 ± 0.8 are consistent with a quadratic-order improvement. They do not exceed the threshold that error-corrected advantage is estimated to require [9], and they are measured at small sizes.

Limitations.

- Instance sizes are limited to $N \leq 20$ by exact diagonalisation.
- Parallel tempering [55] and cluster algorithms [107] were not included, because their gaps are not computable exactly on the extended state space.
- Comparisons count chain steps rather than wall-clock cost, and a quantum step costs more than a classical flip.

- The uniform baseline is idealised.
- Hardware noise was not modelled. The proposal is exactly symmetric only for the ideal circuit, and noise-induced asymmetry would bias the stationary distribution unless mitigated.

Extending the study to $N \geq 30$ by matrix-free Lanczos methods [68], adding parallel tempering by autocorrelation measurements, testing non-stoquastic drivers [23], and costing proposals in physical time on Rydberg hardware [33] would decide whether the Erdős–Rényi scaling persists.

9 Conclusion

Fair allocation of indivisible resources admits an exact reformulation in which fairness is a rule for updating priority weights and computation reduces to Gibbs sampling over the feasible set. This removes the representability obstruction that has shaped quantum treatments of equity. It also shows that fairness itself neither creates nor destroys quantum advantage. For fair scheduling under interference, a blockade-preserving quantum proposal gives an exactly correct sampler with gate-level resources stated in closed form. It yields large constant-factor improvements, lower tracking cost in most instances, and an improvement in gap scaling on dense random graphs, which is instance-dependent rather than general.

Declarations

Data availability. All data are generated by the released code from fixed seeds. The results files are included in the code archive.

Code availability. The complete implementation (fairq, version 1.0.0) with a one-command reproduction script is provided as supplementary material under the MIT licence.

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Author contributions. All authors contributed to the theory, the software, the experiments and the manuscript, and approved the final version.

Use of AI tools. An AI assistant (Claude, Anthropic) was used to support literature search, drafting and code development. All results were generated by the released code, and the authors reviewed the content and take full responsibility for it.

Appendix A. The allowed weights of each criterion

Each entry of Table 2 follows from the conjugate (4).

Max-min. For a probability vector θ , $\theta \cdot y \geq \min_i y_i$, with equality when all y_i are equal, so $W^*(\theta) = 0$. For any other θ the infimum in (4) is $-\infty$: a negative θ_j lets y_j grow without bound, and weights that do not sum to one let a constant vector y run off to $\pm\infty$.

Ordered weighted averages and Gini–Sen welfare. For decreasing ω , the rearrangement inequality [49] gives $\sum_k \omega_k y_{(k)}$ as the minimum of $\sum_i \omega_{\sigma(i)} y_i$ over all permutations σ , so the allowed weights are the permutations of ω and their mixtures. Gini–Sen welfare is the case $\omega_k = (2(n-k)+1)/n^2$, since $\text{mean} \times (1 - \text{Gini}) = n^{-2} \sum_{i,j} \min(y_i, y_j)$ and $y_{(k)}$ is the smaller element of exactly $2(n-k)+1$ ordered pairs.

Nash. $\min_{y>0}(\theta y - \log y) = 1 + \log \theta$, attained at $y = 1/\theta$.

α -fairness. The minimum is attained at $\theta_i = y_i^{-\alpha}$ and gives $W^*(\theta) = \frac{\alpha}{\alpha-1} \sum_i \theta_i^{(\alpha-1)/\alpha}$.

Linear requirements. A requirement that is linear in the lottery, such as a need floor or ex-ante envy-freeness, adds one non-negative multiplier to the weights by Lagrangian duality [16].

Which criteria respect Pareto improvements. A concave W never decreases when some agent gains if and only if all its allowed weights are non-negative [94]. Mean minus variance fails this test: its weights $\theta_i = \frac{1}{n} - \frac{2\lambda}{n}(y_i - \text{mean}(y))$ are negative for agents who are well enough off.

Appendix B. Benchmark instance

The benchmark is `make_instance(20, 3, 0, kind="udg")`. Its edges are

```
(0,5) (0,6) (0,7) (0,8) (0,12) (0,15) (0,17) (0,19) (1,10) (2,3) (2,4) (2,11)
(2,13) (2,14) (2,18) (3,4) (3,8) (3,11) (3,12) (3,14) (3,15) (3,18) (4,11) (4,14)
(4,18) (5,6) (5,7) (6,7) (6,19) (7,12) (7,15) (7,17) (7,19) (8,11) (8,12) (8,14)
(8,15) (8,19) (9,12) (9,16) (9,17) (11,12) (11,14) (11,15) (11,18) (12,14) (12,15)
(12,17) (12,19) (14,15) (14,18) (15,17) (15,19)
```

and its colour classes are $C_1 = \{1, 2, 6, 12, 16\}$, $C_2 = \{4, 5, 9, 10, 13, 15\}$, $C_3 = \{0, 3\}$, $C_4 = \{7, 11\}$, $C_5 = \{14, 17, 19\}$ and $C_6 = \{8, 18\}$. The first driver gate of a step, for vertex 1 with $N(1) = \{10\}$, reads in OpenQASM 2.0 [27]

```
x q[10]; h q[1]; cx q[10],q[1]; rz(-0.1) q[1]; cx q[10],q[1]; rz(0.1) q[1]; h
q[1]; x q[10];
```

The full step is written by `circuit_spec.py` to `results/proposal_step.qasm`, and the rates and owners to `results/circuit_spec.json`.

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